

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

Project Review – 1

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Institution

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Academic Year

2026 – 2027

Problem Statement

Customer churn — the loss of existing customers to competitors or discontinued engagement — is one of the most pressing challenges faced by modern businesses in telecommunications, banking, insurance, subscription services, and e-commerce. Organizations continuously collect large volumes of customer demographic, behavioral, service-usage, transactional, and interaction data, yet identifying which customers are likely to leave remains difficult using conventional reporting methods.

Traditional analysis is largely reactive — churn is identified only after a customer has already left, by which point retention is no longer possible.

There is a clear need for a data-driven predictive system that analyzes customer behavior, uncovers the strongest indicators of churn, and classifies customers into churn / non-churn categories well before attrition occurs.

This project centers on:

- Data Analytics on multi-dimensional customer data
- Exploratory Data Analysis (EDA)
- Classification Modeling for churn prediction
- Feature Engineering from behavioral signals
- Customer Behavior Analysis for retention strategy

Literature Survey

Part 1 of 2

S.No	Title	Source	Contributor
1	Research on Telecom Customer Churn Prediction Based on Ensemble Learning	Journal of Intelligent Information Systems (Springer), 2023	Liu, Y., Fan, J., Zhang, J. et al.
2	SMOTE: Synthetic Minority Over-sampling Technique	Journal of Artificial Intelligence Research, 2002	Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P.
3	A Comparison of Two Oversampling Techniques (SMOTE vs MTDF) for Handling Class Imbalance: A Case Study of Customer Churn Prediction	Springer, 2015	Amin, A., Rahim, F., Ali, I., Khan, C., Anwar, S.
4	Intelligent Data Analysis Approaches to Churn as a Business Problem: A Survey	Knowledge and Information Systems (Springer), 2017	García, D.L., Nebot, À., Vellido, A.
5	A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners	IEEE Access, 2024	Manzoor, A., Qureshi, M.A., Kidney, E., Longo, L.
6	An Efficient Churn Prediction Model Using Gradient Boosting Machine and Metaheuristic Optimization	Scientific Reports, 2023	AlShourbaji, I., Helian, N., Sun, Y., Hussien, A.G., Abualigah, L., Elnaim, B.

Literature Survey

Part 2 of 2

S.No	Title	Source	Contributor
7	Enhancing Bank Customer Churn Prediction from an Imbalanced Dataset Using Machine Learning and SMOTE-Tomek Resampling	Springer (LNNS, COCIA 2025), 2026	Berrada Chakour, O., Maizate, A., Ettaoufik, A., Aissaoui, K.
8	A Survey on Churn Analysis in Various Business Domains	IEEE Access, 2020	Ahn, J., Hwang, J., Kim, D., Choi, H., Kang, S.
9	Arithmetic Optimization with Ensemble Deep Learning SBLSTM-RNN-IGSA Model for Customer Churn Prediction	IEEE Access, 2023	Jajam, N., Challa, N.P., Prasanna, K.S.L., Deepthi, C.H.V.S.
10	Sequential Feature Selection in Customer Churn Prediction Based on Naive Bayes	IOP Conference Series: Materials Science and Engineering, 2020	Yulianti, Y., Saifudin, A.
11	Explainable Artificial Intelligence for Data Science on Customer Churn	IEEE 8th Intl. Conf. on Data Science and Advanced Analytics (DSAA), 2021	Leung, C.K., Pazdor, A.G., Souza, J.

Research Gaps

01

Limited Cross-Model Benchmarking

Few studies evaluate multiple classification models (Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost) on the same churn dataset under identical preprocessing conditions.

02

Shallow Exploratory Analysis

Comprehensive EDA — covering univariate, bivariate, multivariate, and correlation analysis — is often skipped or minimally applied before model development.

03

Class Imbalance Challenges

Churn datasets are inherently imbalanced; many approaches rely on Accuracy alone, understating the importance of Precision, Recall, F1-Score, and ROC-AUC.

04

Narrow Feature Engineering

Limited attention is given to engineering features from tenure, service-usage combinations, billing behavior, and customer support interactions.

05

Lack of Interpretable Dashboards

Practical visualization and analytical dashboards that translate model outcomes into actionable, interpretable churn insights remain scarce.

Objectives

1

Analyze Customer Data

To perform comprehensive Data Analytics and Exploratory Data Analysis on customer data to identify patterns associated with customer churn.

2

Preprocess & Engineer Features

To preprocess customer data and apply feature engineering and feature selection techniques that capture behavioral churn signals.

3

Train Classification Models

To train and compare multiple classification models — Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost — for predicting customer churn.

4

Evaluate & Compare Models

To evaluate and compare the models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and other relevant classification metrics.

Innovation & Novelty

S.No	Innovation	Description
1	Comprehensive Customer Analytics Pipeline	Integrates data preprocessing, extensive EDA, feature engineering, classification modeling, and visualization into a single unified pipeline.
2	Multi-Model Classification Comparison Framework	Compares Logistic Regression, Decision Tree, Random Forest, XGBoost, CatBoost, and Support Vector Machine using identical preprocessing and evaluation criteria.
3	Customer Behavior-Based Feature Engineering	Creates meaningful features from tenure, service usage, contract information, billing behavior, and customer support interactions.
4	Interactive Churn Analytics Dashboard	Displays customer churn patterns, important features, model performance, and high-risk customer insights in one interface.

Project Workflow & Implementation Plan

Part 1 of 2



Project Workflow & Implementation Plan

Part 2 of 2

